

Search with Job-Provided Referrals

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Abstract

This paper explores the impact of job-provided networking and referral opportunities on labor market outcomes. Using data from the Survey of Consumer Expectations, I estimate a continuous-time job search model with bargaining in which workers encounter jobs through either a referral or official channel, and jobs differ in their ability to generate referrals during on-the-job search. The model is estimated separately for workers with and without a college degree. I find that network jobs and referrals support higher wages, with larger wage effects for workers without a college degree. The employment effects depend on how workers respond to having fewer offers. Removing network jobs and referrals lowers employment for college workers, while employment is nearly unchanged for workers without college as they become less selective about the jobs they accept. Across both education samples, job-provided network opportunities are more valuable at low and intermediate match productivity than near the top of the job ladder, because they increase exposure to better, higher paying jobs.

1 Introduction

Referrals and interpersonal networks play an important role in labor market outcomes and are well documented by the surveys of Ioannides and Loury (2004) and Topa (2011). A key finding of this literature is that referral-based search channels shape not only job-finding rates but also the progression of workers over the course of their careers. By transmitting information about vacancies and by providing some form of additional screening, networks affect match quality, job tenure, and the sequence of jobs workers take (Montgomery, 1991; Calvó-Armengol and Jackson, 2004). Access to referral opportunities is not evenly distributed, as some workers are better connected and some jobs provide greater exposure to networks than others. This heterogeneity implies that referrals can affect long-run wage growth and mobility to better jobs, rather than merely improving match quality and wages at referred jobs (Galenianos, 2014).

In terms of empirical results, Brown et al. (2016) use data from a large U.S. firm to show that referred candidates have higher hiring probabilities, initially higher wages, and longer job tenures. Schmutte (2015) uses matched employer–employee data to show that referral networks affect both job assignment and earnings. Similarly, Baranowska-Rataj et al. (2023) finds that low-skilled

workers with high-skilled co-worker networks are more likely to move to better-paid jobs as a result of the network and peer effects.

In looking at the heterogeneous usage and impact of referrals, Loury (2006) finds that referred workers tend to have longer tenures, and attributes these longer tenures to either better matches via the referral channel, or rather, that workers who use referrals tend to have fewer outside options. Using a model with heterogeneous network structures, Arbex et al. (2019) shows that better-connected workers earn higher wages, climb the job ladder more rapidly, and rely more heavily on referrals at higher rungs of that ladder. Lester et al. (2026) further explores heterogeneity along both worker and referral dimensions, splitting the referrals channel into referrals from friends and family and referrals from business contacts. This distinction is important in their calibrated model, as they find referrals from friends and family primarily facilitate employment for workers who struggle to obtain offers through formal channels, while business referrals disproportionately raise wages for workers with already strong labor market opportunities.

This paper contributes to this literature by modeling the impact of work-provided networking opportunities on wages and job-to-job transitions. Using data from the New York Fed’s Survey of Consumer Expectations Job Search Supplement (Armantier et al., 2016), which records referral offers, referral contacts, and referral-based hires for both employed and unemployed individuals, I estimate a model where jobs differ in their ability to generate referral opportunities. This is done by having workers in “network jobs” and “non-network jobs” have different arrival rates for job offers from a referral channel. The objective of this model is to explore how specific jobs provide more access to networking options that can lead to subsequent jobs, and to understand which groups of workers benefit from these jobs the most, and what happens to their wages both at these jobs and subsequent wage growth. The model heavily builds on the structural framework from Lester et al. (2026), but omits worker ability. Workers choose how much effort to put into referral search, given the opportunities their current job provides and what they stand to gain from moving elsewhere. This allows the model to connect the value of a network job to the worker’s position in the job ladder.

2 Model

2.1 Environment

Time is continuous. Workers are risk-neutral and discount at rate $r > 0$. Each job is characterized by a type $n \in \{0, 1\}$ and a match quality $x > 0$. I distinguish non-network jobs, $n = 0$, from network jobs, $n = 1$, according to their ability to generate referral opportunities. An offer is for a job of type n with probability π_n , where $\pi_0 + \pi_1 = 1$. Conditional on job type, match quality is drawn from a lognormal distribution G_n , with $\log x \mid n \sim \mathcal{N}(\mu_n, \sigma_n^2)$. Output is linear in match quality and does not depend directly on job type:

$$f(x) = px + c. \tag{1}$$

Unemployed workers receive flow payoff b . Employed workers experience an exogenous job-destruction rate, $\delta > 0$ that is shared across both network and non-network job types. Productivity

and job type remain fixed within a match. Workers return to unemployment when their job is destroyed and can move directly between employers when they receive an acceptable outside offer.

2.2 Job Arrivals

Workers receive job offers through one of two channels. A referral from a friend, colleague, or business contact arrives through channel R . All other sources of offers are included in the official channel, O . Unemployed workers receive offers through these channels at rates λ_R^U and λ_O^U , giving a total arrival rate

$$\lambda^U = \lambda_O^U + \lambda_R^U. \quad (2)$$

For employed workers, the official offer rate is λ_O^E . Referral arrivals depend on the current job type and the worker's search effort, $s(x, n) \geq 0$:

$$\lambda_R^E(x, n) = \xi_n s(x, n), \quad \lambda^E(x, n) = \lambda_O^E + \lambda_R^E(x, n). \quad (3)$$

Here ξ_n measures how effectively effort produces referral offers in a job of type n . Search has flow cost $C(s) = s^{1+\nu}/(1+\nu)$, where $\nu > 0$. When entering a job, the worker chooses how much effort to put into referral-based search before bargaining over the wage. The wage bargain takes this choice as given, and both search effort and the wage remain fixed for the rest of the match.

Offers from either channel have job type n' with probability $\pi_{n'}$ and match quality x' drawn from $G_{n'}$. The current job affects how often the worker receives referral offers, while the type and quality of offers are drawn in the same way for both channels.

2.3 Value Functions and Wages

Let V^U denote the expected discounted value of unemployment and $V^E(x, n, w)$ the value of working in a job with match quality x , type n , and wage w . Wages are set through Nash bargaining. An unemployed worker bargains against the value of remaining unemployed. For an employed worker, competition between the incumbent and a poaching firm determines the outside option. This three-way bargaining environment follows Dey and Flinn (2005); Cahuc et al. (2006), with the addition of the value of the job-type of the incumbent job and the offered job entering the bargaining.

Acceptance and wage bargaining. An unemployed worker accepts an offer if the job can provide at least the value of unemployment when it pays its full output:

$$V^E(x, n, f(x)) \geq V^U. \quad (4)$$

The reservation match quality x_n^* satisfies $V^E(x_n^*, n, f(x_n^*)) = V^U$ when the threshold is interior. For an accepted offer, the wage $w^u(x, n)$ satisfies

$$V^E(x, n, w^u(x, n)) = V^U + \alpha[V^E(x, n, f(x)) - V^U], \quad (5)$$

where $\alpha \in (0, 1)$ is the worker's share of the surplus over the outside option.

An employed worker in job (x, n) accepts an outside offer (x', n') when

$$V^E(x', n', f(x')) > V^E(x, n, f(x)). \quad (6)$$

Let $A_{n'}(x, n)$ denote the set of offers from a firm of type n' that satisfy this condition. The incumbent can compete for the worker by bidding up to the full output, $f(x)$. The worker's outside option in bargaining with the new firm is therefore $V^E(x, n, f(x))$. The wage at the new employer, $w' = w'(x, n; x', n')$, satisfies

$$\begin{aligned} V^E(x', n', w') &= V^E(x, n, f(x)) \\ &+ \alpha [V^E(x', n', f(x')) - V^E(x, n, f(x))]. \end{aligned} \quad (7)$$

The starting wage consequently depends on both the job the worker leaves and the one they enter. An outside offer that does not lead to a move leaves the existing wage in place.

Value functions. The value of unemployment satisfies

$$rV^U = b + \lambda^U \sum_n \pi_n \int_{x_n^*}^{\infty} [V^E(x, n, w^u(x, n)) - V^U] dG_n(x). \quad (8)$$

The first term is the flow utility from unemployment, b . The second term is the expected gain from accepting an offer at a bargained wage, relative to remaining unemployed, weighted by the offer arrival rate from unemployment, λ^U . Let $\Gamma(x, n)$ be the rate at which an employed worker receives an acceptable outside offer:

$$\Gamma(x, n) = \lambda^E(x, n) \sum_{n'} \pi_{n'} \int_{A_{n'}(x, n)} dG_{n'}(x'). \quad (9)$$

The value of employment then satisfies:

$$\begin{aligned} [r + \delta + \Gamma(x, n)]V^E(x, n, w) &= w - C(s(x, n)) + \delta V^U \\ &+ \lambda^E(x, n) \sum_{n'} \pi_{n'} \int_{A_{n'}(x, n)} V^E(x', n', w') dG_{n'}(x'). \end{aligned} \quad (10)$$

Which represents the value of the current wage net of search costs, with the continuation values following job destruction or a job-to-job move, weighted by their respective transition rates. $r + \delta + \Gamma(x, n)$ combines time discounting with the total rate at which the worker leaves the current job.

Referral search. At hiring, the worker chooses the search plan that maximizes the value obtained in bargaining. Since that value increases with $V^E(x, n, f(x))$, optimal effort satisfies

$$s(x, n)^\nu = \alpha \xi_n \sum_{n'} \pi_{n'} \int_{A_{n'}(x, n)} [V^E(x', n', f(x')) - V^E(x, n, f(x))] dG_{n'}(x'). \quad (11)$$

Where the left-hand side is the marginal cost of on-the-job search effort, while the right-hand side measures the expected benefit from additional referral offers, accounting for the worker's bargaining share. Within each job type, workers in more valuable matches have less to gain from moving and therefore choose lower search effort. The value of a job thus depends on its current wage and the opportunities it provides to move to better jobs.

3 Estimation

3.1 Data

To estimate the model, I use data from the Job Search Supplement (JSS) from the Survey of Consumer Expectations (SCE) (Armantier et al., 2016). The JSS was administered annually between 2013 and 2024 and includes a repeated cross-section of just over 13,000 U.S. adults' labor-force status, job search behavior, job offers, referral channels, and wages. The survey asks employed workers to report the channel through which they received their current job, including separate categories for referrals from "a friend or relative," "a former co-worker, supervisor, or business associate," and "a current employee at the company". The survey also asks about ongoing job search activity for both employed and unemployed individuals by inquiring about the number of offers received in the last four weeks and last six months. I will use the information on the frequency and sources of these offers to calibrate the job arrival rates by channel. I will split the sample into those without a college degree and those with at least an undergraduate degree.

3.2 Empirical Motivation

To illustrate the role of referrals in job search, I first look at how often workers use them to obtain a job and how often they receive new contacts through them. Table 1 reports these measures separately for referrals from friends or relatives, former colleagues or associates, and employees of the hiring or prospective firm. About 40% of current jobs were obtained through a referral. And 8.0% of employed workers and 17.9% of unemployed workers report having received a referral contact within the last four weeks prior to responding to the JSS.

The sources of referrals differ across education groups. Workers without a college degree are more likely to have found their current job through a friend or relative compared to workers with a college degree. College graduates are more likely to have been referred by a former colleague or associate, or by an employee of the hiring firm. This pattern is consistent with Lester et al. (2026), who find that referrals from family and friends are more common in lower-skill occupations, while business referrals are more common in higher-skill occupations. Recent contacts show a similar difference for friends and relatives, with higher shares among workers without a college degree in both employment states. Contacts from former colleagues or associates are slightly more common among college graduates.

The question for the model is whether some jobs provide more access to these subsequent referral opportunities than others. I compare the new referrals workers receive according to how they obtained their current job. I call a job a *network job* if the worker reports having been referred to it, and a *regular job* if the worker reports no hiring referral. A worker can receive new referrals in either type of job, and the source of a new referral can differ from the source that helped them obtain their current job.

For this comparison, I restrict the employed sample to respondents with known hiring sources and at least three months of current-job tenure. This places the hiring event before the period in which recent contacts are measured. Workers who report no employer contacts are counted as

Table 1: Prevalence of Referrals in Hiring and Job Search

Referral source	All	College+	Less than College
<i>A. Current job obtained through a referral</i>			
Any referral source	40.2 (0.58)	38.4 (0.72)	43.1 (0.95)
Friend or relative	25.6 (0.51)	22.2 (0.62)	31.2 (0.89)
Former colleague / associate	12.4 (0.39)	14.1 (0.52)	9.6 (0.57)
Employee of hiring firm	11.3 (0.37)	12.0 (0.48)	10.1 (0.58)
<i>Observations</i>	7,241	4,527	2,714
<i>B. Recent referral contacts: employed</i>			
Any referral source	8.0 (0.32)	7.8 (0.40)	8.2 (0.52)
Friend or relative	3.0 (0.20)	2.2 (0.22)	4.2 (0.38)
Former colleague / associate	4.5 (0.24)	4.6 (0.31)	4.2 (0.38)
Employee of prospective firm	3.3 (0.21)	3.2 (0.26)	3.6 (0.35)
<i>Observations</i>	7,354	4,572	2,782
<i>C. Recent referral contacts: unemployed</i>			
Any referral source	17.9 (1.70)	16.2 (2.26)	19.8 (2.56)
Friend or relative	9.1 (1.27)	7.5 (1.62)	10.7 (1.99)
Former colleague / associate	8.5 (1.23)	9.4 (1.79)	7.4 (1.69)
Employee of prospective firm	8.3 (1.22)	7.1 (1.58)	9.5 (1.89)
<i>Observations</i>	508	266	242

Notes: Standard errors in parentheses. Panel A reports the percentage of employed workers who obtained their current job through each referral source. Panels B and C report the percentage of employed and unemployed workers, respectively, contacted by a potential employer in the last four weeks as a result of a referral. These contacts need not necessarily lead to job offers.

having no referral contacts, while missing responses are excluded. Each comparison includes all workers with valid responses in the relevant job type, including those who received no referral.

Recent referrals in regular and network jobs. Figure 1 compares the referral contacts received by workers in regular and network jobs. In both education groups, workers in network jobs report a higher incidence of new referral contacts. The figure also separates these contacts by source, showing how the difference between the two job types varies across the connections through which workers receive referrals. Each source refers to the recent contact; a worker who originally obtained a job through a friend can subsequently receive a referral from a former colleague.

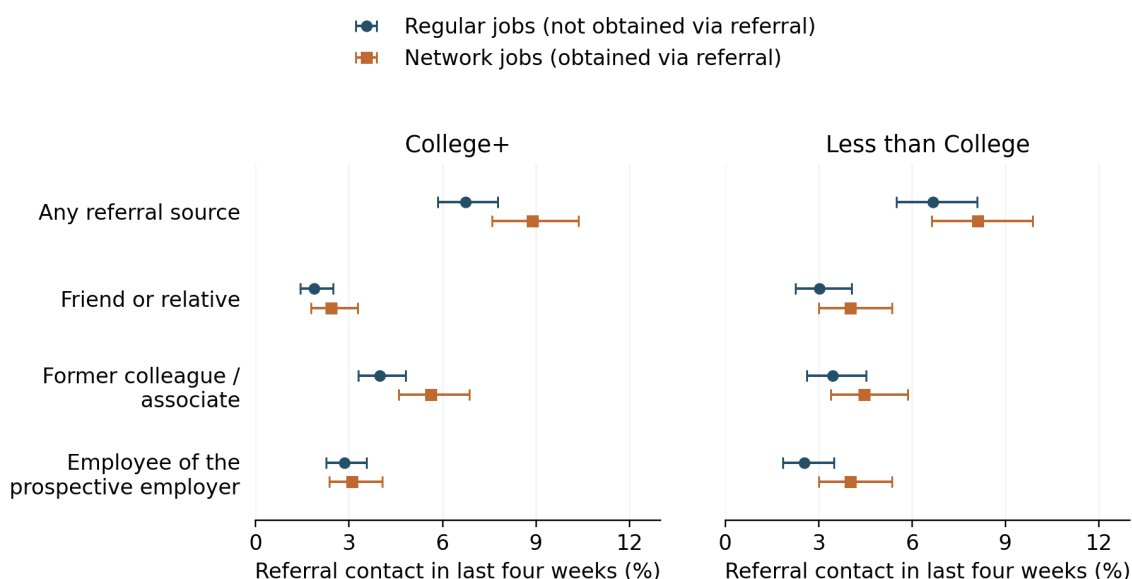


Figure 1: Recent Referral Contacts in Regular and Network Jobs, by Contact Source

Notes: Points show the percent receiving a referral contact in the last four weeks; bars are 95 percent confidence intervals. Regular jobs were not obtained through a reported referral; network jobs were obtained through at least one reported referral. Rows refer to the source of the recent contact, which can differ from the original hiring source. Current-job tenure is at least three months.

The difference in contact rates between network jobs and regular jobs suggest that the connections associated with entering a job may continue to matter for subsequent job search, or that job type provides additional opportunities for referrals. However, it's important to note that a worker who takes a network job as opposed to a regular job probably differs in their access to social and professional networks outside of those provided by the job, so this comparison cannot alone identify how much network jobs contribute to increased referral access.

Recent referrals over job tenure. Figure 2 shows how referral contacts vary with time in the current job. The blue line shows workers in regular jobs (those not obtained via referrals) and the orange line shows workers who originally obtained their jobs through a referral. In both cases, the groups are defined by the hiring channel, so receiving a new referral does not change a worker's

classification.

Referral contacts are generally less common among workers who have been in their jobs longer, although the decline is not uniform across the tenure bins. One possible explanation is that workers in better matches have less reason to search and are also less likely to leave. These workers would become more common among the jobs observed at long tenures, creating a decline in average contacts even if search effort is fixed within each match. The figure compares different workers at different tenures, so it is consistent with this kind of selection. I use the relationship to help estimate search behavior in the model.

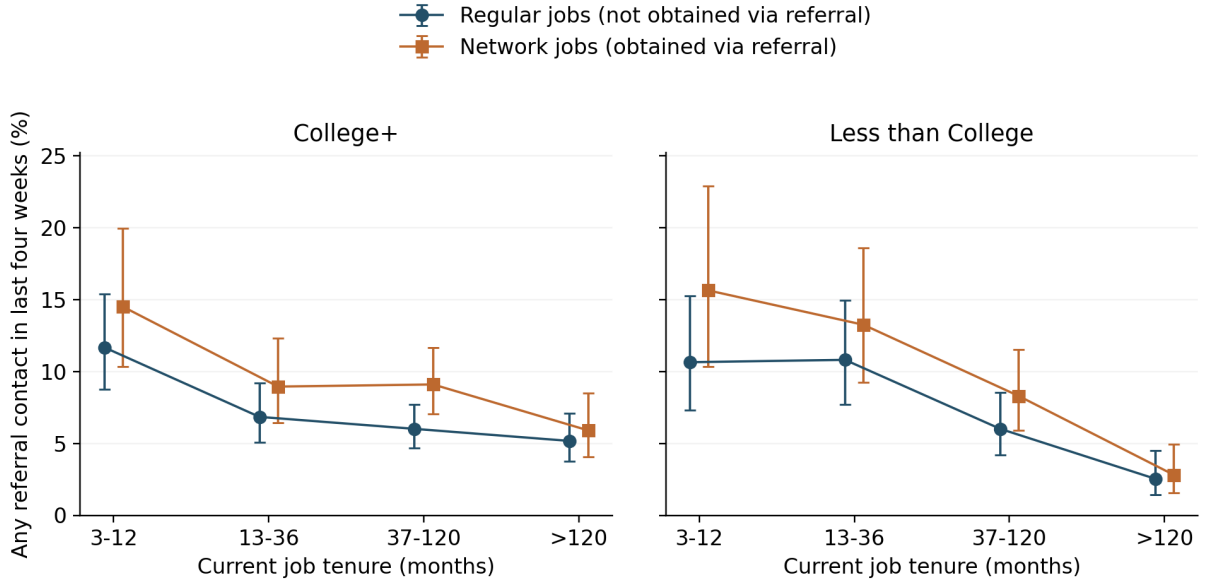


Figure 2: Recent Referral Contacts by Job Tenure: Regular versus Network Jobs

Notes: Shares receiving at least one referral contact in the last four weeks. 95 percent confidence intervals. Contacts include all three referral sources. The sample is employed respondents with known hiring sources and at least three months of current-job tenure.

3.3 Assumptions

To simplify estimation, I fix the worker's bargaining weight, $\alpha = 0.5$, which provides the worker with half the surplus generated relative to their outside option. Time is measured in months, and the annual discount rate is 4 percent, so $r = 0.04/12$. I normalize the median productivity of non-network offers to 0.1, or $\mu_0 = \log(0.1)$, to fix productivity units. The output slope p is estimated. The largest difficulty with estimating the model from the JSS data is identifying job type. Since the networking capacity provided by the job is rather intangible, the type of the job must be determined by a variable, or a series of covariates, in the data. I set a job type to $n = 1$ if the worker was themselves referred to that job by a friend or relative, a former co-worker, supervisor or business associate, or a current employee at the company. This follows the referral indicator used to construct the empirical moments. The rationale is that obtaining a job through a referral may signal access to networks that also provide subsequent opportunities. This is far from the ideal way

to identify a network job, as a better way to identify these jobs is by some measurement of how client-facing the role is, whether or not the job pays for conference attendance, or the proportion of past employees that were poached from the firm via referrals. However, given the variables I have from the JSS, I choose to identify the type of a job based on the channel through which the worker received the job offer.

3.4 Method of Moments

Since the JSS asks workers about offers and sources, I calibrate the unemployed official and referral arrival inputs and the employed official input directly from the survey responses.¹ For employed workers, the model determines referral arrivals through search effort and the opportunities provided by the current job. I estimate the referral efficiencies ξ_0 and ξ_1 using the contact-by-tenure moments. These moments help distinguish a job that makes referrals easier to obtain from a worker's incentive to keep looking. I also estimate a common effort-cost curvature, ν , for both job types within each education group. The cost scale is normalized to one because a change in effort units could otherwise be absorbed into the referral efficiencies.

The estimated parameters are

$$\theta = (\sigma_0, \mu_1, \sigma_1, \pi_1, p, c, \delta, b, \xi_0, \xi_1, \nu). \quad (12)$$

I estimate these eleven parameters using twenty-six moments. For each candidate vector, I solve the value functions and stationary distribution and compute the corresponding model moments $\widehat{m}(\theta)$. With $\ell(\theta) = \widehat{m}(\theta) - m^{\text{data}}$, the estimator is

$$\widehat{\theta} = \arg \min_{\theta \in \Theta} \ell(\theta)^\top W \ell(\theta), \quad (13)$$

where the diagonal entries of W are inverse bootstrap variances of the data moments.

I match unemployment and the share of employed workers in network jobs, along with the EU and EE targets. The wage and job-tenure moments help determine the kinds of matches workers accept and how long they remain in them. The wage moments are fractions below the tenth, twenty-fifth, fiftieth, seventy-fifth, and ninetieth data percentiles within each job type. For the tenure moments, let $e_n(x, \tau)$ be the stationary density of workers in jobs of type n , match quality x , and tenure τ , integrated over wages. If $e_n(x, 0)$ is the flow of workers entering these jobs, then

$$e_n(x, \tau) = e_n(x, 0) e^{-[\delta + \Gamma(x, n)]\tau}. \quad (14)$$

The tenure moments are fractions with current-job tenure of one year or less and three years or less:

$$\Pr(T \leq t \mid E, n) = \frac{\int_0^\infty \int_0^t e_n(x, \tau) d\tau dx}{\int_0^\infty \int_0^\infty e_n(x, \tau) d\tau dx}, \quad t \in \{12, 36\}. \quad (15)$$

¹These offer inputs combine the four-week and six-month questions and use the reported offer source. The unemployment inputs use the nonemployment sample, which differs from the actively searching or temporarily laid-off sample in Table 1. The EU target is the sample share currently jobless after a recent involuntary separation; the EE target is the sample share reporting acceptance of an outside offer received while employed. I use the survey offer and contact incidences as inputs for the corresponding monthly model rates.

Eight additional moments describe referral contacts by job type and tenure: 3–12 months, 13–36 months, 37–120 months, and more than 120 months. For a bin $[\tau_L, \tau_H]$, the model counterpart is

$$\frac{\int_{\tau_L}^{\tau_H} \int e_n(x, \tau) \lambda_R^E(x, n) dx d\tau}{\int_{\tau_L}^{\tau_H} \int e_n(x, \tau) dx d\tau}. \quad (16)$$

Since a referral contact does not necessarily lead to a job offer. I use the share of workers receiving referral contacts as a measure of their access to referral offers, without separately estimating how often these contacts turn into offers.

Tables 2 and 3 compare the model and data moments. Table 4 reports the parameters and calibrated arrival inputs. Figure 3 compares the wage distributions in the model and data, and Figure 4 shows the fitted contact profiles. The model captures the broad decline in referral contacts and the wage distributions, but understates the EU and EE targets for both education groups.

Standard errors come from 200 respondent bootstrap samples within each education group. Each sample recomputes the data moments, directly calibrated arrival inputs, and re-estimates all eleven parameters. Wage thresholds and the weighting matrix remain at their full-sample values. Reported standard errors are the standard deviations of the resulting estimates. All three counterfactuals are re-solved for every bootstrap estimate.

Table 2: Employment, Wage, and Tenure Moments

Moment	College+		Less than College	
	Model	Data	Model	Data
Unemployment rate	0.038	0.042	0.059	0.062
Network-job share among employed	0.340	0.340	0.363	0.364
EU target	0.006	0.010	0.006	0.013
EE target	0.013	0.030	0.011	0.020
Non-network wages below data p10	0.095	0.100	0.102	0.100
Non-network wages below data p25	0.255	0.250	0.248	0.250
Non-network wages below data p50	0.497	0.500	0.501	0.500
Non-network wages below data p75	0.749	0.750	0.755	0.750
Non-network wages below data p90	0.899	0.900	0.901	0.900
Network wages below data p10	0.115	0.100	0.102	0.100
Network wages below data p25	0.254	0.250	0.250	0.250
Network wages below data p50	0.487	0.500	0.497	0.500
Network wages below data p75	0.755	0.750	0.742	0.750
Network wages below data p90	0.902	0.900	0.900	0.900
Non-network tenure: 1 year or less	0.179	0.177	0.178	0.203
Non-network tenure: 3 years or less	0.393	0.371	0.396	0.375
Network tenure: 1 year or less	0.177	0.181	0.159	0.161
Network tenure: 3 years or less	0.392	0.398	0.369	0.332

Table 3: Referral Contact Moments by Job Tenure

Moment	College+		Less than College	
	Model	Data	Model	Data
Non-network contacts: 3–12 months	0.115	0.117	0.145	0.106
Non-network contacts: 13–36 months	0.093	0.068	0.104	0.108
Non-network contacts: 37–120 months	0.066	0.060	0.057	0.060
Non-network contacts: over 120 months	0.041	0.052	0.025	0.025
Network contacts: 3–12 months	0.126	0.145	0.157	0.156
Network contacts: 13–36 months	0.102	0.089	0.113	0.132
Network contacts: 37–120 months	0.073	0.091	0.065	0.083
Network contacts: over 120 months	0.046	0.059	0.030	0.028

Table 4: Estimated Model Parameters by Education Group

	College+	Less than College
b	5.34 (1.90)	2.74 (1.03)
p	12.46 (1.35)	5.28 (0.60)
c	8.94 (0.71)	6.17 (0.52)
δ	0.0058 (0.0002)	0.0064 (0.0003)
λ_O^U	0.205 (0.027)	0.272 (0.028)
λ_R^U	0.112 (0.020)	0.140 (0.022)
λ_O^E	0.154 (0.005)	0.167 (0.006)
μ_0	-2.303 (fixed)	-2.303 (fixed)
σ_0	1.500 (0.039)	1.575 (0.042)
μ_1	-1.386 (0.094)	-0.073 (0.119)
σ_1	1.300 (0.047)	1.066 (0.059)
π_1	0.181 (0.012)	0.069 (0.008)
ξ_0	0.043 (0.004)	0.053 (0.003)
ξ_1	0.046 (0.005)	0.058 (0.004)
ν	0.766 (0.188)	0.360 (0.048)

Notes: Standard errors from 200 respondent bootstrap re-estimates per education group are in parentheses. The three exogenous arrival inputs are recalculated in each sample. Time is measured in months; $r = 0.04/12$ and $\alpha = 0.5$. Effort costs are normalized to $C(s) = s^{1+\nu}/(1+\nu)$. The last three rows report referral efficiencies and the common cost curvature; employed referral arrivals are endogenous.

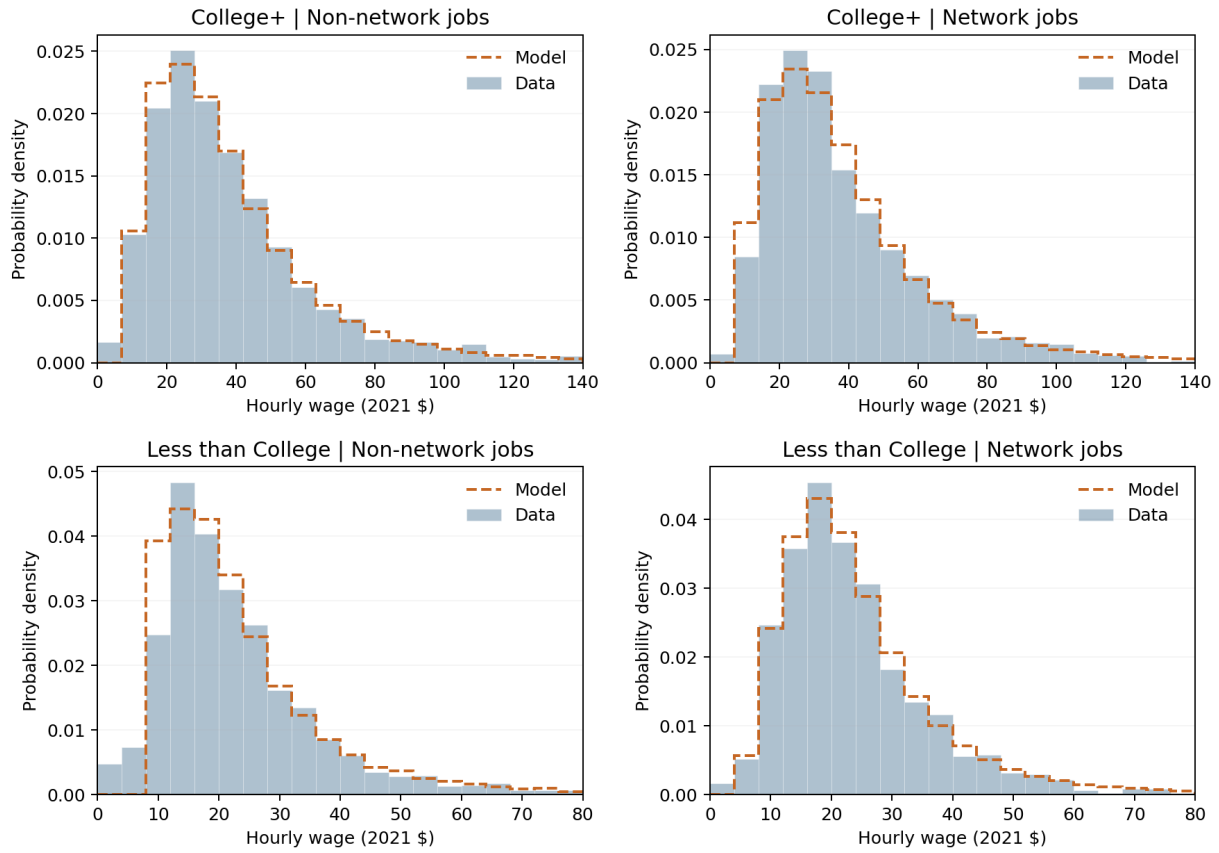


Figure 3: Hourly Wage Distributions in the Model and Data

Notes: Bars show the data and the dashed line shows the model, using common wage bins. Wages are in 2021 dollars.

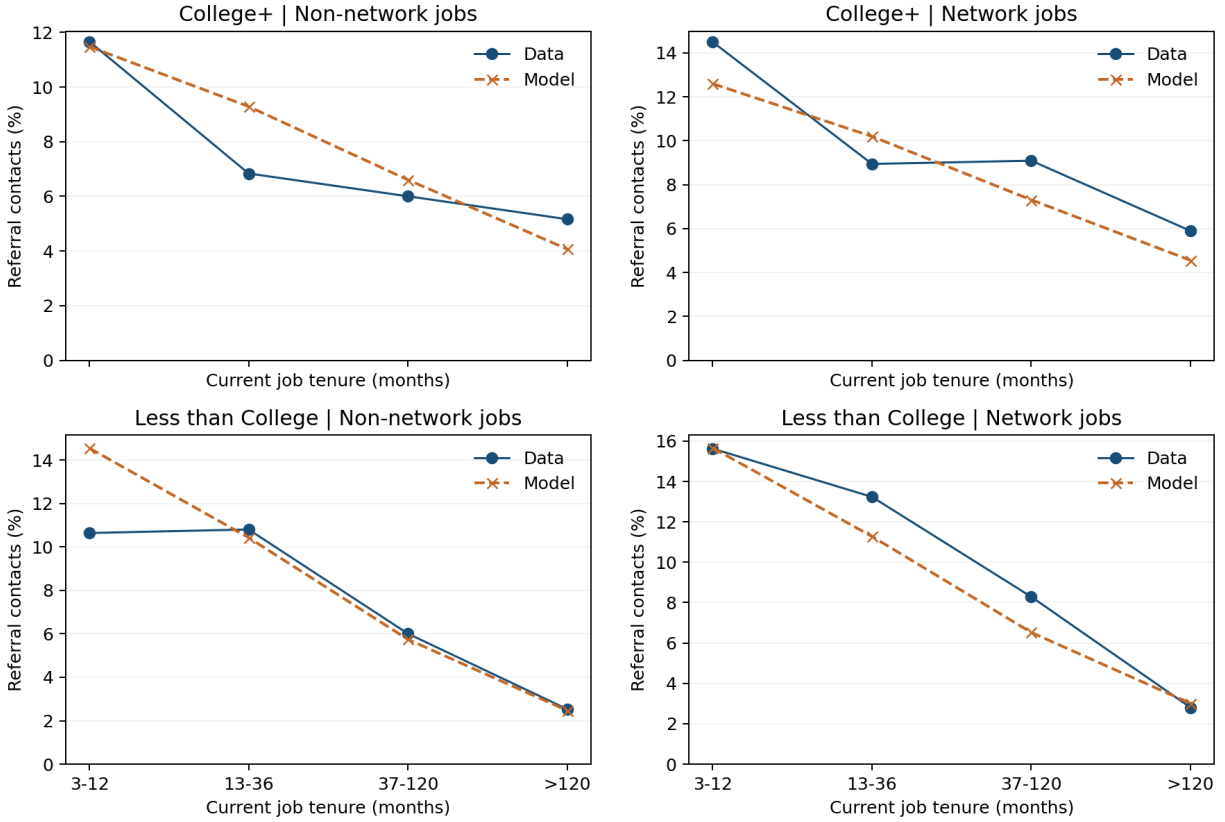


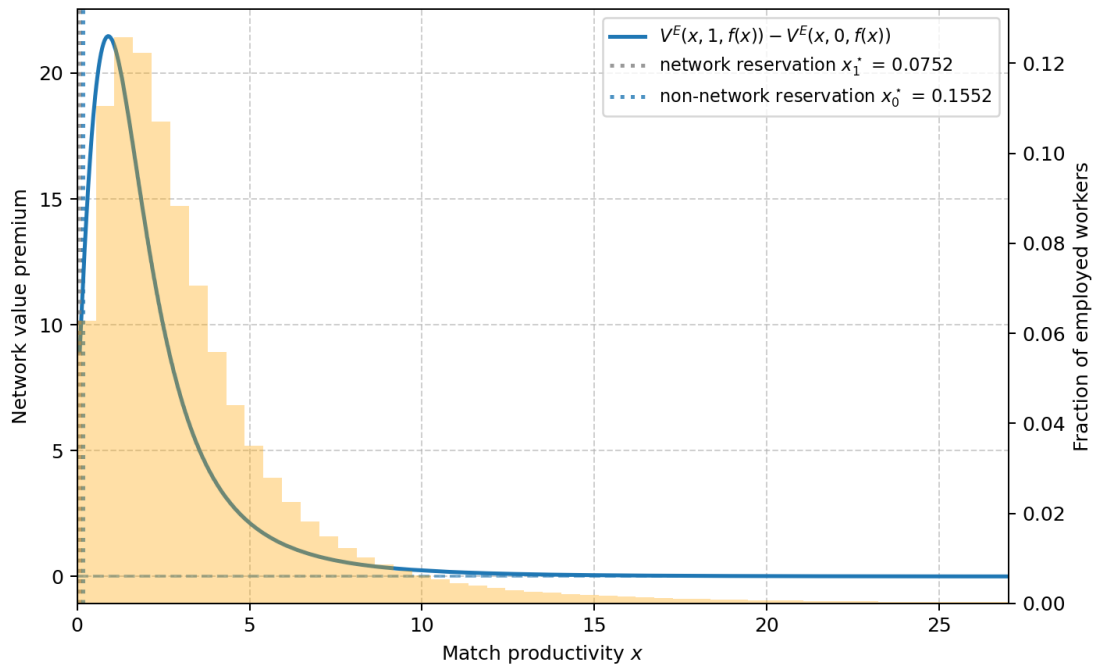
Figure 4: Referral Contact Profiles in the Model and Data

Notes: Points show the data moments and crosses show the corresponding model moments within job type and tenure group.

4 Analysis of Calibrated Model

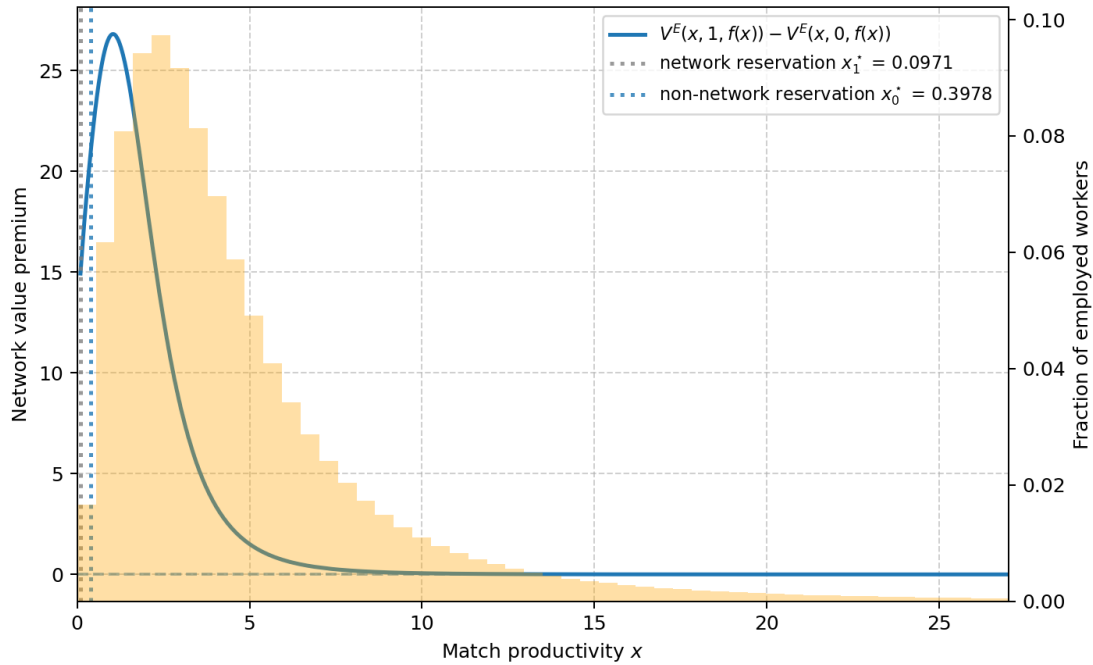
A key feature of the model is that workers lower on the wage ladder benefit more from network jobs because referrals provide opportunities to move to better-paying jobs. For workers already higher on the ladder, fewer outside offers are worth accepting, so access to referrals provides less of a benefit. The additional value of being in a network job therefore decreases as workers move up the wage ladder. To illustrate this difference, Figures 5 and 6 compare the value of network and non-network jobs at the same match productivity for the College+ and Less than College samples. Both jobs are evaluated at a wage equal to output, so the difference reflects the opportunities the jobs provide and the effort workers choose to put into search. As the figures show, the network premium is largest toward the lower end of the productivity distribution and becomes small among the most productive matches. The orange histograms show that many employed workers are in the range where these opportunities are valuable. This relates to the finding of Baranowska-Rataj et al. (2023) that coworker networks can help low-wage workers move into better-paid jobs.

Figure 5: College+: Value Function Premium for a Network Job



Notes: The blue line is $V^E(x, 1, f(x)) - V^E(x, 0, f(x))$. Dotted lines mark entry reservation productivities. The orange histogram shows the stationary distribution of employed workers over match productivity on the right axis.

Figure 6: Less than College: Value Function Premium for a Network Job



Notes: The blue line is $V^E(x, 1, f(x)) - V^E(x, 0, f(x))$. Dotted lines mark entry reservation productivities. The orange histogram shows the stationary distribution of employed workers over match productivity on the right axis.

Search effort also helps explain why referral contacts decline with tenure in Figure 4. Workers in better matches have less to gain from moving to another job and therefore choose to search less. These matches are also more likely to last, as fewer outside offers are worth accepting. Workers with longer tenure are therefore more likely to be in good matches where they have less reason to search, leading to fewer referral contacts on average.

To quantify the role of these opportunities, I first re-solve the model giving network jobs the same ability to generate referrals as non-network jobs, while keeping the productivity distributions and the types of jobs being offered the same. This asks how much of the benefit of a network job comes from the additional access to referrals it provides. The results in Table 5 show that employment and median wages fall in both education groups, although the wage changes are relatively modest.

I next remove network jobs entirely, setting $\pi_1 = 0$. Workers now draw all offers from the non-network productivity distribution, so they lose access to both the matches and the referral opportunities associated with network jobs. The wage decline is much larger in this exercise, particularly for the Less than College group. This suggests that the kinds of matches these jobs provide are an important part of their contribution to wages, alongside the opportunities they create for subsequent moves.

As a final exercise, I remove the referral channel as well, so unemployed and employed workers can receive offers only through the official channel. Median wages fall further in both groups.

Workers have fewer opportunities to move on from low-paying jobs, and an unemployed worker deciding whether to accept an offer has less reason to wait for a better one. Throughout these exercises, I hold official arrival rates fixed and allow search effort, job acceptance and bargained wages to adjust.

The effect on employment depends on how workers adjust the jobs they are willing to accept. When network jobs and referrals are removed, unemployed workers in both education groups accept a larger share of the offers they receive. For College+ workers, this is not enough to offset the lower offer rate, so employment falls. For Less than College workers, the two effects almost cancel out, leaving employment nearly unchanged. Wages still fall substantially in both groups, as workers have fewer opportunities to move to better-paying jobs.

Table 5: Contribution of Network Jobs and Referrals to Wages and Employment

Outcome	Exercise	College+	Less than College
Employment rate	Benchmark	0.962 (0.008)	0.941 (0.009)
	Equal referral technology	-0.31% (0.16)	-0.40% (0.14)
	No network jobs	-1.30% (0.71)	-2.68% (1.08)
	No network jobs or referrals	-1.15% (1.67)	0.04% (2.02)
Median wage	Benchmark	\$32.72 (0.32)	\$20.46 (0.20)
	Equal referral technology	-0.44% (0.31)	-1.15% (0.52)
	No network jobs	-11.04% (1.03)	-20.41% (0.98)
	No network jobs or referrals	-30.11% (1.89)	-35.83% (1.99)

Notes: Standard errors from 200 bootstrap re-estimates per education group are in parentheses. Counterfactual entries are percent changes from the benchmark. Equal referral technology sets $\xi_1 = \xi_0$ and re-solves effort, preserving both productivity distributions, job-type offer probabilities, and unemployed arrivals. No network jobs sets $\pi_1 = 0$. No network jobs or referrals additionally removes unemployed referral arrivals and both employed referral efficiencies. Official arrivals and all other parameters stay fixed.

5 Concluding Remarks

The results of the estimated model show how network jobs can affect a worker's earnings through the opportunities they provide for future employment. These opportunities are especially valuable to workers in less productive matches, since they have more to gain from moving to another job. The wage effects of removing network jobs and referrals are larger for the Less than College group, suggesting that access to these opportunities plays a greater role in supporting their wages. However, the employment response also depends on what workers are willing to accept when offers become harder to obtain. For workers without college, the greater willingness to accept offers almost offsets the loss of referrals, leaving employment nearly unchanged while wages fall. Employment falls for college workers because their increase in acceptance is not enough to offset the reduction in offers.

There are several limitations to this interpretation. I use the source through which a worker obtained their current job as a measure of access to job-provided networking. Some of the differences

in subsequent referrals may instead come from the worker's own connections or from the kinds of workers who obtain these jobs; so some measure of workers' abilities and existing networks should be added. The worker-specific network could be modeled dynamically, where workers are able to keep some of the connections they develop when moving between jobs. On the demand side, firms could also choose how much access to networks they provide. These connections may help a firm recruit new workers, but they may also make its current workers more likely to leave, or decrease output in some way. Allowing firms to make this choice would help explain why some jobs provide more networking opportunities than others. Linked worker-employer data would help distinguish connections developed in the current job from those brought from previous jobs. Another extension would be to add general equilibrium, allowing firms to adjust vacancy creation in response to changes in referrals. This would allow the model to account for how firms' hiring decisions change the opportunities available to workers and, in turn, affect wages and employment.

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